

BLM5106- Advanced Algorithm Analysis and Design

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[Introduction to algorithms](#) TH Cormen, CE Leiserson, RL Rivest, C Stein

Hash Functions

A hash function h maps arbitrary strings of data to fixed length output. The function is deterministic and public, but the mapping should look “random”. In other words,

$$h : \{0, 1\}^* \rightarrow \{0, 1\}^d$$

for a fixed d . Hash functions do not have a secret key. Since there are no secrets and the function itself is public, anyone can evaluate the function. To list some examples,

Hash function	MD4	MD5	SHA-1	SHA-256	SHA-512
d	128	128	160	256	512

Hash Functions

- In practice, hash functions are used for “digesting” large data. For example, if you want to check the validity of a large file (potentially much larger than a few megabytes), you can check the hash value of that file with the expected hash.
- That is, it should be “hard” to find two inputs m_1 and m_2 for hash function h such that $h(m_1) = h(m_2)$.

Desirable Properties

- One-way (pre-image resistance)
- Collision-resistance (Strong collision-resistance)
- Target collision resistance (Weak collision-resistance)
- Pseudo-random
- Non-malleability

One-way (pre-image resistance)

- Given $y \in \{0,1\}^d$, it is hard to find an x such that $h(x) = y$.
- Assume $h(x) = x \bmod p$, is $h(x)$ one way?

Collision-resistance

- It is hard to find any pair of inputs x, x' such that $x \neq x'$ and $h(x) = h(x')$
- Is it important for storing passwords?
- What about a look up table?

Target collision resistance

- Given x , it is hard to find x' such that $x \neq x'$ and $h(x) = h(x')$

Pseudo-random

- The function behaves indistinguishable from a random oracle.
- The Random Oracle model is an ideal model of the hash function that is not achievable in practice.
- In this model, we assume there exists an oracle h such that on input $x \in \{0,1\}^*$, if h has not seen x before, then it outputs a random value as $h(x)$. Otherwise, it returns $h(x)$ it previously output.
- Unfortunately, a random oracle does not exist since it requires infinite storage, so in practice we use pseudo-random functions.

Non-malleability

- Given $h(x)$, it is hard to generate $h(x')$ where x and x' are related
exp: $x' = x + 1$

Are these properties imply others?

- Collision resistance (CR) $\rightarrow$ Target collision resistance (TCR) (but not reverse)
- if h is OW is it also CR and TCR?
- if h is TCR or CR, is it also OW?

CRIPTOGTAFIC HASH FUNCTION APPLICATIONS

- Password Storage
- File Modificatition Detector
- Digital Signature
- Commitment

Password Storage

- We can store hash $h(p)$ for password p instead of p directly, and check $h(p)$ to authenticate a user.
- If it satisfies the property **OW**, adversary comprising $h(p)$ will not learn p .
- What about CR and TCR?

File Modification Detector

- For each file F , we can store $h(F)$ in a secure location. To check authenticity of a file, we can recompute $h(F)$.
- What is a successful break?
 - Adversary want to modify F to F' but keep $h(F)=h(F')$
- This requires property TCR.
- What about OW?

Digital Signature

- We can use hash functions to generate a signature that guarantees that the message came from a said source.
- Signing: $\partial = \text{sign}(\text{SK}, M)$
- Verificaiton : $\text{Verify}(M, \partial, \text{PK}) = \text{True/False}$

M, H(M) OW?

Commitment

- In a secure bidding, Alice wants to bid value x , but does not want to reveal the bid until the auction is over.
- Alice then computes $C(x)$, and publicize it, which serves as her commitment.
- When bidding is over, then she can reveal x , and x can be verified using $C(x)$.

Commitment

- $C(x)$:
 - It must not reveal X (OW)
 - It must also protect system from Alice (CR)
 - Given $C(X)$ should not be possible to produce $C(X+1)$ (NM)